

8.11

Parasitology

1. **The *incorrect* match between organism and the appropriate diagnostic procedure is:**
 - A. *Onchocerca volvulus*—examination of skin snips
 - B. *Cryptosporidium* spp.—modified acid-fast stain
 - C. *Enterobius vermicularis*—routine ova and parasite examination
 - D. *Schistosoma haematobium*—examination of urine sediment

Microbiology/Apply knowledge of diagnostic techniques/Parasitology/2
2. **In a patient with diarrhea, occasionally *Entamoeba histolytica*/*E. dispar* (four nucleated cysts, no chromatoidal bars) are identified as being present; however, these cells, which are misdiagnosed as protozoa, are really:**
 - A. Macrophages
 - B. Polymorphonuclear leukocytes (PMNs)
 - C. Epithelial cells
 - D. Eosinophils

Microbiology/Apply knowledge of the morphology of artifacts/Parasitology/3
3. **Charcot-Leyden crystals in stool may be associated with an immune response and are thought to be formed from the breakdown products of:**
 - A. Neutrophils
 - B. Eosinophils
 - C. Monocytes
 - D. Lymphocytes

Microbiology/Apply knowledge of the morphology of artifacts/Parasitology/1
4. **Parasitic organisms that may be transmitted sexually *do not* include:**
 - A. *Entamoeba gingivalis*
 - B. *Dientamoeba fragilis*
 - C. *Trichomonas vaginalis*
 - D. *Diphyllobothrium latum*

Microbiology/Apply knowledge of life cycles and epidemiology/Parasitology/1

Answers to Questions 1–4

1. **C** The appropriate procedure for the diagnosis of *E. vermicularis* would involve the microscopic examination of a series of consecutive cellulose tapes. Typical eggs and/or a rare adult pinworm could be identified under the microscope.
2. **B** As PMNs in stool begin to fragment and appear to have four nuclei, they will resemble *E. histolytica*/*E. dispar* cysts. However, *E. histolytica*/*E. dispar* cysts are rarely seen in cases of diarrhea. The species name *E. histolytica* is reserved for the true pathogen, whereas *E. dispar* is used for the nonpathogenic species. Unfortunately, they look identical morphologically. The only time *E. histolytica* could be identified

morphologically would be from trophozoites containing ingested RBCs. Nonpathogenic *E. dispar* would not contain ingested RBCs. The correct way to report these organisms is *E. histolytica/E. dispar* (no trophozoites containing ingested RBCs) or *E. histolytica* (trophozoites seen that contain ingested RBCs). Physicians may base the treatment on patient symptoms.

3. **B** When eosinophils disintegrate, the granules reform into Charcot-Leyden crystals.
 4. **D** Of the organisms listed, *Diphyllbothrium latum* cannot be transmitted sexually; this infection is acquired from the ingestion of infected raw or poorly cooked freshwater fish.
5. **The incorrect match between the organism and one method of acquiring the infection is:**
- A. *Trypanosoma brucei rhodesiense*—bite of sand fleas
 - B. *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*)—ingestion of water contaminated with cysts
 - C. Hookworm—skin penetration of larvae from soil
 - D. *Toxoplasma gondii*—ingestion of infected raw or rare meats
- Microbiology/Apply knowledge of fundamental life cycles/Parasitology/1
6. **Upon examination of stool material for *Cystoisospora belli*, one would expect to see:**
- A. Cysts containing sporozoites
 - B. Precysts containing chromatoidal bars
 - C. Oocysts that are modified acid-fast positive
 - D. Sporozoites that are hematoxylin positive
- Microbiology/Apply knowledge of life cycles and organism morphology/Parasitology/1
7. **Which specimen is the least likely to provide recovery of *Trichomonas vaginalis*?**
- A. Urine
 - B. Urethral discharge
 - C. Vaginal discharge
 - D. Feces
- Microbiology/Apply knowledge of pathogenesis and diagnostic procedures/Parasitology/2
8. **Which of the following is the best technique to identify *Dientamoeba fragilis* in stool?**
- A. Formalin concentrate
 - B. Trichrome-stained smear
 - C. Modified acid fast–stained smear
 - D. Giemsa stain
- Microbiology/Apply knowledge of diagnostic procedures/Parasitology/2
9. **One of the following protozoan organisms has been implicated in waterborne and foodborne outbreaks within the United States. The suspect organism is:**
- A. *Pentatrichomonas hominis*
 - B. *Dientamoeba fragilis*
 - C. *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*)
 - D. *Balantidium coli*
- Microbiology/Apply knowledge of life cycles and epidemiology/Parasitology/1
10. **Gram staining of a sample from a gum lesion showed what appeared to be amoebae. A trichrome smear showed amoebae with a single nucleus and partially digested PMNs.**

The correct identification is:

- A. *Trichomonas tenax*
- B. *Entamoeba histolytica*/*Entamoeba dispar*
- C. *Entamoeba gingivalis*
- D. *Entamoeba polecki*

Microbiology/Apply knowledge of organism morphology and body site/Parasitology/3

Answers to Questions 5–10

- 5. **A** East and West African trypanosomiasis (*T. B. rhodesiense* and *T. B. gambiense*) are acquired when infective forms are introduced into the human body through the bite of the tse-tse fly, not sand fleas.
 - 6. **C** *C. belli* oocysts in various stages of maturity would be seen in the concentrate sediment or possibly the direct, wet preparation; these oocysts would stain positive with modified acid-fast stains.
 - 7. **D** *T. vaginalis* is site-specific. The organisms are found in the urogenital tract; thus, the intestinal tract is not the normal site for these organisms.
 - 8. **B** The best technique to recover and identify *D. fragilis* trophozoites and rare cysts would be the trichrome-stained smear.
 - 9. **C** For a number of years, *G. lamblia* (*G. intestinalis*, *G. duodenalis*) has been implicated in both waterborne and foodborne outbreaks from the ingestion of infective cysts within contaminated water and food.
 - 10. **C** *E. gingivalis* is known to be an inhabitant of the mouth and is characterized by morphology that resembles *E. histolytica*/*E. dispar*. However, *E. gingivalis* tends to ingest PMNs, whereas *E. histolytica*/*E. dispar* do not.
- 11. An *E. histolytica* (true pathogen) trophozoite has the following characteristics:**
- A. Compact karyosome in the nucleus, ingested RBCs, and clear pseudopodia
 - B. Ingested RBCs, clear pseudopodia, and uneven chromatin on the nuclear membrane
 - C. Ingested RBCs, clear pseudopodia, and large glycogen vacuoles in cytoplasm
 - D. Large, blotlike karyosome, ingested white blood cells (WBCs), and granular pseudopods
- 12. A 12-year-old girl is brought to the emergency room with meningitis and a history of swimming in a warm-water spring. Motile amoebae that measure 10 µm in size are seen in CSF and are most likely:**
- A. *Iodamoeba bütschlii* trophozoites
 - B. *Endolimax nana* trophozoites
 - C. *Dientamoeba fragilis* trophozoites
 - D. *Naegleria fowleri* trophozoites
- Microbiology/Apply knowledge of life cycle and epidemiology/Parasitology/3
- 13. Characteristics of the rhabditiform (noninfective) larvae of *Strongyloides stercoralis* include a:**
- A. Short buccal capsule and large genital primordium
 - B. Long buccal capsule and pointed tail
 - C. Short buccal capsule and small genital primordium

D. Small genital primordium and notch in tail

Microbiology/Apply knowledge of organism morphology and life cycle/Parasitology/2

14. Visceral larva migrans is associated with which of the following organisms and diagnostic method?

A. *Toxocara*—serology

B. *Onchocerca*—skin snips

C. *Dracunculus*—skin biopsy

D. *Angiostrongylus*—CSF examination

Microbiology/Apply knowledge of life cycle and diagnostic procedures/Parasitology/2

15. The following organisms are linked with specific, relevant information. The incorrect combination is:

A. *Strongyloides stercoralis*—internal autoinfection

B. *Echinococcus granulosus*—hydatid examination

C. *Toxoplasma gondii*—serology

D. *Balantidium coli*—common in the United States

Microbiology/Apply knowledge of life cycle and epidemiology/Parasitology/2

16. Examination of 24-hour unpreserved urine specimen is sometimes helpful in the recovery of:

A. *Trichomonas vaginalis* trophozoites

B. *Schistosoma haematobium* eggs

C. *Enterobius vermicularis* eggs

D. *Strongyloides stercoralis* larvae

Microbiology/Apply knowledge of life cycle and diagnostic methods/Parasitology/1

Answers to Questions 11–16

11. **A** The trophozoite of *E. histolytica* has evenly arranged chromatin on the nuclear membrane; a central, compact karyosome in the nucleus; clear pseudopodia; and ingested RBCs in the cytoplasm.
12. **D** *N. fowleri* are free-living soil and water amoebae that cause primary amebic meningoencephalitis (PAM). The number of cases reported is few; however, the infection is very acute and almost always fatal.
13. **A** The rhabditiform larvae of *S. stercoralis* are characterized by the short buccal capsule (mouth) and large genital primordium, whereas hookworm larvae have a long buccal capsule and very small genital primordium.
14. **A** *Toxocara* spp. are the cause of visceral larva migrans and occur when humans accidentally ingest the infective eggs of the dog or cat ascarid. The larvae migrate through the deep tissues, including the eye. The test of choice is the serology.
15. **D** *B. coli* is a ciliate that can cause watery diarrhea in humans; however, it is not commonly found within the United States. It is the largest of the intestinal protozoa and can be found in proficiency testing specimens. Therefore, although it is not common, laboratories must still be able to identify these organisms.
16. **B** *S. haematobium* blood flukes reside in the veins over the bladder. When the eggs are

passed from the body, they are often found in urine; egg viability can also be determined in unpreserved urine.

17. The examination of sputum may be necessary to diagnose infection with:

- A. *Paragonimus westermani*
- B. *Trichinella spiralis*
- C. *Wuchereria bancrofti*
- D. *Fasciola hepatica*

Microbiology/Apply knowledge of life cycle and diagnostic methods/Parasitology/1

18. Two helminth eggs that may resemble one another are:

- A. *Diphyllobothrium latum* and *Paragonimus westermani*
- B. *Opisthorchis sinensis* and *Fasciolopsis buski*
- C. *Taenia saginata* and *Hymenolepis nana*
- D. *Ascaris lumbricoides* and *Trichostrongylus*

Microbiology/Apply knowledge of organism morphology/Parasitology/2

19. Eating poorly cooked infected pork can lead to an infection with:

- A. *Taenia solium* and *Trichinella spiralis*
- B. *Taenia saginata* and *Hymenolepis nana*
- C. *Trichuris trichiura* and *Hymenolepis diminuta*
- D. *Diphyllobothrium latum* and *Ascaris lumbricoides*

Microbiology/Apply knowledge of organism life cycle/Parasitology/1

20. An operculated cestode egg that can be recovered from human feces is:

- A. *Clonorchis sinensis*
- B. *Diphyllobothrium latum*
- C. *Paragonimus westermani*
- D. *Dipylidium caninum*

Microbiology/Apply knowledge of organism morphology/Parasitology/1

21. The adult tapeworm of *Echinococcus granulosus* is found in the intestine of:

- A. Dogs
- B. Sheep
- C. Humans
- D. Cattle

Microbiology/Apply knowledge of life cycle/Parasitology/1

22. In infections with *Taenia solium*, humans can serve as the:

- A. Definitive host
- B. Intermediate host
- C. Either the definitive or the intermediate host
- D. None of the above

Microbiology/Apply knowledge of life cycle/Parasitology/2

23. Humans acquire infections with *Diphyllobothrium latum* adult worms by:

- A. Ingestion of freshwater crabs
- B. Skin penetration of cercariae
- C. Ingestion of water chestnuts
- D. Ingestion of raw freshwater fish

Microbiology/Apply knowledge of life cycle/Parasitology/1

Answers to Questions 17–23

17. **A** *P. westermani* adult worms are found in the lung, and eggs may be coughed up in the sputum. Consequently, both sputum and stool (if the sputum containing the eggs is swallowed) are the recommended specimens for examination for the eggs.
18. **A** Both *D. latum* and *P. westermani* eggs are operculated and approximately the same size. The morphology is similar, although *D. latum* has a knob at the abopercular end and *P. westermani* has a thickened abopercular end and shoulders into which the operculum fits.
19. **A** Both *T. solium* (pork tapeworm) and *T. spiralis* can be acquired from the ingestion of raw or poorly cooked infected pork.
20. **B** *D. latum* is the only operculated cestode egg that is found in humans; the infection is acquired from the ingestion of infected raw or poorly cooked freshwater fish.
21. **A** Although hydatid cysts are found in sheep or in humans (accidental intermediate host), the adult tapeworms of *E. granulosus* are found in the intestine of the dog.
22. **C** If humans ingest *T. solium* cysticerci in uncooked or rare pork, the adult tapeworm will mature within the intestine (human will serve as definitive host); if eggs from the adult tapeworm are ingested, then the cysticerci will develop in human tissues (accidental intermediate host), causing cysticercosis.
23. **D** The ingestion of raw freshwater fish containing the encysted larvae of *D. latum* will result in the development of an adult tapeworm within the human intestine.

24. Humans can serve as both the intermediate and definitive host in infections caused by:

- A. *Enterobius vermicularis*
- B. *Hymenolepis nana*
- C. *Schistosoma japonicum*
- D. *Ascaris lumbricoides*

Microbiology/Apply knowledge of life cycle/Parasitology/1

25. *Babesia* is an organism that has been implicated in disease from both splenectomized and nonsplenectomized patients. Morphologically, the parasites resemble:

- A. *Plasmodium falciparum* rings
- B. *Leishmania donovani* amastigotes
- C. *Trypanosoma cruzi* trypomastigotes
- D. Microsporidian spores

Microbiology/Apply knowledge of parasite morphology/Parasitology/2

26. Organisms (and infections) that under normal conditions *cannot* be transmitted in the laboratory are:

- A. *Cryptosporidium*—cryptosporidiosis
- B. *Taenia solium*—cysticercosis
- C. *Ascaris lumbricoides*—ascariasis
- D. *Enterobius vermicularis*—pinworm infections

Microbiology/Apply knowledge of life cycles/Parasitology/2

27. *Toxoplasma gondii* is characterized by:

- A. Possible congenital infection and ingestion of oocysts
- B. Cosmopolitan distribution and possible difficulties with interpretation of serological results
- C. None of the above
- D. Both A and B

Microbiology/Apply knowledge of all areas of parasite biology, diagnostic procedures/Parasitology/3

28. Oocysts of *Cryptosporidium* spp. can be detected in stool specimens by using:

- A. Modified acid-fast stain
- B. Gram stain
- C. Methenamine silver stain
- D. Trichrome stain

Microbiology/Apply knowledge of diagnostic procedures, staining characteristics/Parasitology/1

29. Which microfilariae are usually *not found* circulating in peripheral blood?

- A. *Brugia malayi*
- B. *Wuchereria bancrofti*
- C. *Onchocerca volvulus*
- D. *Loa loa*

Microbiology/Apply knowledge of diagnostic procedures, staining characteristics/Parasitology/1

30. Massive hemolysis, blackwater fever, and central nervous system involvement are *most common with*:

- A. *Plasmodium vivax*
- B. *Plasmodium falciparum*
- C. *Plasmodium ovale*
- D. *Plasmodium malariae*

Microbiology/Apply knowledge of disease pathogenesis/Parasitology/2

Answers to Questions 24–30

- 24. **B** In infections with *H. nana*, humans serve as both intermediate and definitive hosts. When ingested, the oncosphere penetrates the intestinal mucosa, develops into the mature cysticercoid (human is intermediate host), and returns to the gut, where the adult tapeworm matures (human is definitive host). Thus, the life cycle can go from egg to adult tapeworm within the human host.
- 25. **A** *Babesia* is an intracellular parasite that closely resembles the ring forms (early trophozoites) of *P. falciparum*. Often in babesiosis there are more rings per cell and the ring form is the only stage seen.
- 26. **C** *A. lumbricoides* eggs require a period of development in the soil before they are infective for humans. The other organisms listed can be transmitted within the laboratory or in the hospital setting.
- 27. **D** Infection with *T. gondii* is acquired through the ingestion of infected rare or raw meats, infective oocysts from cat feces, or as a congenital transmission. The organism

has a cosmopolitan distribution and although serological testing is generally the test of choice, the results may be very difficult to interpret in certain situations (e.g., congenital infection and immunocompromised patients).

28. **A** Oocysts of *Cryptosporidium* spp. can be found and identified using microscopic examination of fecal smears stained with modified acid-fast stains. They appear as purple-red-pink round objects, measuring approximately 4 to 6 μm . Often the four sporozoites and residual body can be seen within the oocyst wall.
29. **C** *O. volvulus* microfilariae are normally found in the fluid right under the outer layer of skin. Therefore, the skin snip is the proper specimen to examine.
30. **B** The pathogenic sequelae of malarial infections with *P. falciparum* are the most severe of the four species listed. They can include massive hemolysis, blackwater fever, and multiple organ involvement, including the central nervous system (cerebral malaria).

31. Organisms that should be considered in a nursery school outbreak of diarrhea include:

- A.** *Endolimax nana*, *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*), and *Entamoeba coli*
B. *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*), *Dientamoeba fragilis*, and *Cryptosporidium parvum*
C. *Cryptosporidium* spp., *Trichomonas vaginalis*, and *Entamoeba coli*
D. *Trichomonas hominis*, *Dientamoeba fragilis*, and *Endolimax nana*
- Microbiology/Apply knowledge of epidemiology/Parasitology/2*

32. The incorrect match between disease and symptoms is:

- A.** Paragonimiasis—hemoptysis
B. Cryptosporidiosis—watery diarrhea
C. Toxoplasmosis in compromised host—central nervous system symptoms
D. Enterobiasis—dysentery

Microbiology/Apply knowledge of life cycles/Parasitology/2

33. The formalin-ether (ethyl acetate) concentration procedure for feces is used to demonstrate:

- A.** Motility of helminth larvae
B. Protozoan cysts and helminth eggs
C. Formation of amoebic pseudopods
D. Trophozoites

Microbiology/Apply knowledge of diagnostic procedures/Parasitology/2

34. Cysts of *Iodamoeba bütschlii* typically have:

- A.** Chromatoidal bars with rounded ends
B. A heavily vacuolated cytoplasm
C. A large glycogen vacuole
D. Many ingested bacteria and yeast cells

Microbiology/Apply knowledge of morphology/Parasitology/1

35. The miracidial hatching test helps to demonstrate the viability of eggs of:

- A.** *Taenia* species
B. *Schistosoma* species
C. Hookworm species

D. *Opisthorchis* species

Microbiology/Apply knowledge of diagnostic procedures/Parasitology/1

36. Organisms that should be considered in a waterborne outbreak of diarrheal disease include:

A. *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*) and *Cryptosporidium* spp.

B. *Endolimax nana* and *Entamoeba histolytica*

C. *Blastocystis* spp. and *Trichomonas vaginalis*

D. *Toxoplasma gondii* and *Schistosoma mansoni*

Microbiology/Apply knowledge of epidemiology/Parasitology/2

37. Fecal immunoassays have become more commonly used to diagnose infections with:

A. *Endolimax nana* and *Blastocystis* spp.

B. *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*) and *Cryptosporidium* spp.

C. *Ascaris lumbricoides* and *Trichuris trichiura*

D. *Strongyloides stercoralis* and *Trichomonas vaginalis*

Microbiology/Apply knowledge of pathogenesis and diagnostic procedures/Parasitology/3

Answers to Questions 31–37

- 31. B** *G. lamblia* (*G. intestinalis*, *G. duodenalis*), *D. fragilis*, and *Cryptosporidium* have been implicated in nursery school outbreaks. Among the many protozoa and coccidia found in the human, these three organisms have become the most likely parasites in this type of setting.
- 32. D** Infections with *E. vermicularis* (the pinworm) may cause anal itching, sleeplessness, and possibly some vaginal irritation or discharge; however, dysentery (bloody diarrhea) has not been associated with this infection.
- 33. B** The ova and parasite examination contains three components: the direct wet film (demonstrates protozoan trophozoite motility), the formalin-ethyl acetate concentration (demonstrates protozoan cysts, coccidian oocysts, and helminth eggs), and the trichrome or iron hematoxylin–stained smear (confirms protozoan cysts and trophozoites).
- 34. C** The cyst of *I. bütschlii* is characterized by a large glycogen vacuole that is seen on the wet smear (stains brown with iodine) and on the permanent stained smear (vacuole will appear clear). Occasionally the vacuole will be so large that the organism will collapse on itself.
- 35. B** The determination of egg viability is important in schistosomiasis; therefore, the miracidial hatching test is helpful in demonstrating the egg viability of *Schistosoma* species. Once the eggs are hatched, the living miracidium larvae will be visible in the water.
- 36. A** Both *G. lamblia* (*G. intestinalis*, *G. duodenalis*) and *Cryptosporidium* have been implicated in waterborne outbreaks or diarrheal disease. These infections would result from the ingestion of *Giardia* cysts and/or *Cryptosporidium* oocysts.
- 37. B** Fecal immunoassays have become more commonly used for the diagnosis of infections with *Giardia* and *Cryptosporidium*. These tests are available in EIA, FA, or

rapid cartridge formats.

38. The most common infections with the microsporidia often originate in:

- A. The lung
- B. The nervous system
- C. The gastrointestinal tract
- D. Mucocutaneous lesions

Microbiology/Apply knowledge of life cycles/Parasitology/2

39. Eye infections with *Acanthamoeba* spp. have most commonly been traced to:

- A. Use of soft contact lenses
- B. Use of hard contact lenses
- C. Use of contaminated lens care solutions
- D. Failure to remove lenses while swimming

Microbiology/Apply knowledge of epidemiology/Parasitology/2

40. Select the *most sensitive recovery method for Acanthamoeba* spp. from lens care solutions or corneal biopsies.

- A. The trichrome staining method
- B. The use of monoclonal reagents for the detection of antibody
- C. The use of non-nutrient agar cultures seeded with *Escherichia coli*
- D. The Giemsa stain method

Microbiology/Apply knowledge of diagnostic procedures/Parasitology/2

41. The microsporidia are organisms (now classified with the fungi) that have been implicated in human disease primarily in:

- A. Immunocompromised patients
- B. Pediatric patients under age 5 years
- C. Adult patients with congenital immunodeficiencies
- D. Patients who have been traveling in the tropics

Microbiology/Apply knowledge of pathogenesis and epidemiology/Parasitology/2

42. When staining *Cystoisospora belli* oocysts with modified acid-fast stains, the important difference between these methods and the acid-fast stains used for acid-fast bacilli (AFB) is:

- A. The staining time is much longer with regular AFB acid-fast stains.
- B. The decolorizer is much weaker than acid alcohol used for AFB decolorizing.
- C. A counterstain must be used for the modified methods.
- D. The stain is more concentrated when staining for AFB.

Microbiology/Apply knowledge of diagnostic procedures/Parasitology/2

43. The *incorrect* match between symptoms and disease is:

- A. Dysentery and amebiasis
- B. Malabsorption syndrome and giardiasis
- C. Cardiac involvement and chronic Chagas disease
- D. Myalgias and trichuriasis

Microbiology/Apply knowledge of life cycle and pathogenesis/Parasitology/2

38. C With the possible exception of direct inoculation infection in the eye, the microsporidia are thought to initially infect the gastrointestinal (GI) tract through ingestion of the infective spores; infections in other body sites are thought to disseminate from the GI tract.
39. C The majority of eye infections with *Acanthamoeba* spp. have resulted from the use of contaminated eye care solutions, primarily the use of homemade saline. It is recommended that all solutions be discarded at the expiration date. Continued use may increase the risk of environmental contamination of the fluids.
40. C Currently, the most sensitive method of those listed for the recovery of *Acanthamoeba* spp. from clinical specimens is the non-nutrient agar culture plate seeded with *E. coli*. The amoebae feed on the bacteria; both trophozoites and cysts can be recovered from the agar surface.
41. A Although the microsporidia have been known as pathogens in many groups of animals, their involvement in humans has primarily been in immunocompromised patients, especially those with AIDS. Microsporidia can be found in different tissues, and currently, there are a number of genera implicated in human disease.
42. B The decolorizer in modified acid-fast stains (Kinyoun cold method, modified hot method) is usually 1% to 3% sulfuric acid rather than the much stronger acid alcohol used in the routine AFB stains.
43. D *T. trichiura* (whipworm) may cause diarrhea and occasionally dysentery in very heavy infections; however, the worms are confined to the intestine, and myalgias are not seen in this helminth infection.

44. The incorrect match between organism and characteristic is:

- A. *Chilomastix mesnili* and Shepherd's crook and lemon shape
- B. *Plasmodium malariae* and "band trophozoite"
- C. *Hymenolepis nana* and striated shell
- D. *Wuchereria bancrofti* and sheathed microfilariae

Microbiology/Apply knowledge of morphology/Parasitology/2

45. The incorrect match between method and method objective is:

- A. Direct wet examination and detection of organism motility
- B. Knott concentration and the recovery of operculated helminth eggs
- C. Baermann concentration and the recovery of *Strongyloides*
- D. Permanent stained fecal smear and confirmation of protozoa

Microbiology/Apply knowledge of diagnostic procedures/Parasitology/2

46. The incorrect match between organism and characteristic/diagnostic method is:

- A. *Dientamoeba fragilis* and tetrad karyosome in the nucleus
- B. *Toxoplasma gondii* and diagnostic serology
- C. *Echinococcus granulosus* and daughter cysts
- D. *Schistosoma mansoni* and egg with terminal spine

Microbiology/Apply knowledge of morphology/Parasitology/2

47. There are few procedures considered STAT in parasitology. The most obvious situation would be:

- A. Ova and parasite examination for giardiasis

- B. Baermann concentration for strongyloidiasis
- C. Blood films for malaria
- D. Culture of amebic keratitis

Microbiology/Apply knowledge of pathogenesis and diagnostic procedures/Parasitology/3

48. An immunosuppressed man has several episodes of pneumonia, intestinal pain, sepsis with gram-negative rods, and a history of military service in Southeast Asia 20 years earlier. The most likely cause is infection with:

- A. *Trypanosoma cruzi*
- B. *Strongyloides stercoralis*
- C. *Naegleria fowleri*
- D. *Paragonimus westermani*

Microbiology/Apply knowledge of pathogenesis and life cycles/Parasitology/3

49. In a pediatric patient, the recommended clinical specimens for recovery of *Enterobius vermicularis* are the:

- A. Stool specimens
- B. Sigmoidoscopy scrapings
- C. Duodenal aspirates
- D. Series of cellophane tape preparations

Microbiology/Apply knowledge of pathogenesis and life cycle/Parasitology/2

Answers to Questions 44–49

- 44. C** *H. nana* has a thin eggshell containing a six-hooked embryo (oncosphere) and polar filaments that lie between the eggshell and the embryo. The striated eggshell is generally associated with the eggs of *Taenia* spp.
- 45. B** The Knott concentration is designed to allow the recovery of microfilariae from a blood specimen. Dilute formalin (2%) is used; blood is introduced into the formalin, the red cells lyse, and the sediment can be examined as a wet preparation or permanent stained smear (Giemsa or hematoxylin-based stain) for the presence of microfilariae.
- 46. D** The egg of *S. mansoni* is characterized by a large lateral spine; *S. haematobium* has the characteristic terminal spine.
- 47. C** The request for blood films for malaria should always be considered a STAT request. Any laboratory providing these services should have coverage 24 hours a day, 7 days a week. In cases of *P. falciparum* malaria, any delay in diagnosing the infection could be fatal for the patient.
- 48. B** A latent infection with *S. stercoralis* acquired years before may cause severe symptoms in the immunosuppressed patient (“autoinfective” capability of life cycle and migratory route through the body leading to the hyperinfection syndrome and disseminated disease).
- 49. D** The collection of a series of consecutive Scotch/cellophane tape preparations is recommended for the diagnosis of infection with the pinworm *E. vermicularis*.

50. Eosinophilic meningoencephalitis is a form of larva migrans, causing fever, headache, stiff neck, and increased cells in the spinal fluid. It is caused by:

- A. *Necator americanus*
- B. *Angiostrongylus cantonensis*
- C. *Ancylostoma braziliense*
- D. *Strongyloides stercoralis*

Microbiology/Apply knowledge of pathogenesis and life cycle/Parasitology/2

51. **“Cultures of parasites are different from bacterial cultures; no quality control is needed.” This statement is:**

- A. True, if two tubes of media are set up for each patient
- B. True, if the media are checked every 24 hours
- C. False, unless two different types of media are used
- D. False, and organism and media controls need to be set up

Microbiology/Apply knowledge of diagnostic procedures/Parasitology/2

52. **Protozoan cysts were seen in a concentration sediment and tentatively identified as *Entamoeba coli*. However, the organisms were barely visible on the permanent stained smear because:**

- A. The organisms were actually not present in the concentration sediment
- B. There were too few cysts to allow identification on the stained smear
- C. *Entamoeba coli* cysts were present but poorly fixed due to the thick cyst wall
- D. The concentrate and permanent stained smear were not from the same patient

Microbiology/Apply knowledge of fixatives and diagnostic procedures/Parasitology/3

53. **When humans have hydatid disease, the causative agent and host classification are:**

- A. *Echinococcus granulosus*—accidental intermediate host
- B. *Echinococcus granulosus*—definitive host
- C. *Taenia solium*—accidental intermediate host
- D. *Taenia solium*—definitive host

Microbiology/Apply knowledge of life cycles/Parasitology/2

54. **A 45-year-old hunter developed fever, myalgia, and periorbital edema. He has a history of bear meat consumption. The most likely causative agent is:**

- A. *Toxoplasma gondii*
- B. *Taenia solium*
- C. *Hymenolepis nana*
- D. *Trichinella spiralis*

Microbiology/Apply knowledge of pathogenesis and life cycles/Parasitology/3

55. **In a condition resulting from the accidental ingestion of eggs, the human becomes the intermediate host, rather than the definitive host. The correct answer is:**

- A. Trichinosis
- B. Cysticercosis
- C. Ascariasis
- D. Strongyloidiasis

Microbiology/Apply knowledge of pathogenesis and life cycles/Parasitology/3

Answers to Questions 50–55

50. **B** Eosinophilic meningoencephalitis is a form of larva migrans and is caused by *A. cantonensis*, the rat lungworm. This infection, common in the Pacific region, is

associated with CSF symptoms and sometimes eye involvement.

51. **D** Duplicate cultures should be set up, and specific American Type Culture Collection (ATCC) strains should be cultured along with the patient specimens to confirm that the culture system is operating properly. This approach is somewhat different from that used in diagnostic bacteriology and mycology.
52. **C** As *E. coli* cysts mature, the cyst wall becomes more impenetrable to fixatives. Consequently, the cysts may be visible in the concentrate sediment but appear very distorted or pale on the permanent stained smear.
53. **A** The cause of hydatid disease is *E. granulosus*, and the human is classified as the accidental intermediate host. Infection occurs when humans accidentally ingest the eggs of *E. granulosus* and the hydatid cysts develop in the liver, lung, and other organs of the human instead of sheep (normal cycle).
54. **D** Bear meat is another excellent source of *T. spiralis*. In this case, the patient had evidently consumed poorly cooked infected bear meat, thus ingesting the encysted larvae of *T. spiralis*.
55. **B** The accidental ingestion of *T. solium* eggs can result in the disease called *cysticercosis*. The cysticerci will develop in a number of different tissues, including the brain, and humans are accidental intermediate hosts.

56. **A transplant recipient on immunosuppressive drugs developed increasing diarrhea. The most likely combination of disease and diagnostic procedure is:**

- A. Trichinosis and trichrome staining
- B. Microsporidiosis and modified trichrome staining
- C. Toxoplasmosis and Gram staining
- D. Paragonimiasis and wet preparation

Microbiology/Apply knowledge of pathogenesis and diagnostic procedures/Parasitology/3

57. **After returning from a 2-year stay in India, a patient has eosinophilia, an enlarged left spermatic cord, and bilateral inguinal lymphadenopathy. The most likely clinical specimen and organism match is:**

- A. Thin blood films and *Leishmania*
- B. Urine and concentration for *Trichomonas vaginalis*
- C. Thin blood films and *Babesia*
- D. Thick blood films and microfilariae

Microbiology/Apply knowledge of pathogenesis and diagnostic procedures/Parasitology/3

58. **Patients with severe diarrhea should use “enteric precautions” to prevent nosocomial infections with:**

- A. *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*)
- B. *Ascaris lumbricoides*
- C. *Cryptosporidium* spp.
- D. *Cystoisospora belli*

Microbiology/Apply knowledge of pathogenesis and life cycles/Parasitology/3

59. **A 60-year-old Brazilian patient with cardiac irregularities and congestive heart failure suddenly dies. Examination of the myocardium revealed numerous amastigotes, an indication that the cause of death was most likely:**

- A. Leishmaniasis with *Leishmania donovani*
- B. Leishmaniasis with *Leishmania braziliense*
- C. Trypanosomiasis with *Trypanosoma gambiense*
- D. Trypanosomiasis with *Trypanosoma cruzi*

Microbiology/Apply knowledge of pathogenesis and life cycles/Parasitology/3

60. When analysis of blood smears for malaria are requested, what patient information should be obtained?

- A. Diet, age, gender
- B. Age, antimalarial medication, gender
- C. Travel history, antimalarial medication, date of return to United States
- D. Fever patterns, travel history, diet

Microbiology/Apply knowledge of pathogenesis and life cycle, and epidemiology/Parasitology/3

61. In an outbreak of diarrheal disease traced to a municipal water supply, the most likely causative agent is:

- A. *Cryptosporidium* spp.
- B. *Cystoisospora belli*
- C. *Entamoeba histolytica*
- D. *Dientamoeba fragilis*

Microbiology/Apply knowledge of life cycles and epidemiology/Parasitology/2

Answers to Questions 56–61

- 56. B** The fact that the patient has received a transplant, is on immunosuppressive drugs, and has increasing diarrhea, suggests infection with microsporidia; the most appropriate method would be the modified trichrome stain.
- 57. D** On the basis of the history, the most relevant procedure to perform is the preparation and examination of thick blood films for the recovery and identification of microfilariae. The symptoms suggest early filariasis.
- 58. C** *Cryptosporidium* oocysts (unlike those of *C. belli*) are immediately infective when passed in stool, and nosocomial infections have been well documented with this coccidian.
- 59. D** *T. cruzi*, the cause of Chagas disease, exists in two forms within humans, the trypomastigote in the blood and the amastigote in the striated muscle (usually cardiac muscle and intestinal tract muscle).
- 60. C** Travel history (areas of drug resistance), the date of return to the United States (primary versus relapse case), and history of antimalarial medication and illness (severe illness, few organisms on smear) are very important questions to ask. Without this information, a malaria diagnosis can be missed or delayed with severe patient consequences.
- 61. A** *Cryptosporidium* can be transmitted through contaminated municipal water sources. Such outbreaks have been well documented.

62. Within the United States, sporadic mini-outbreaks of diarrheal disease have been

associated with the ingestion of strawberries, raspberries, fresh basil, mesclun (baby lettuce leaves), and snow peas. The *most* likely causative agent is:

- A. *Dientamoeba fragilis*
- B. *Cyclospora cayetanensis*
- C. *Schistosoma mansoni*
- D. *Cystoisospora belli*

Microbiology/Apply knowledge of life cycles and epidemiology/Parasitology/2

63. Which of the following statements is true regarding onchocerciasis?

- A. The adult worm is present in blood
- B. The microfilariae are in blood during the late evening hours
- C. The diagnostic test of choice is the skin snip
- D. The parasite resides in the deep lymphatics

Microbiology/Apply knowledge of life cycles and diagnostic procedures/Parasitology/2

64. The *most* prevalent helminth to infect humans is:

- A. *Enterobius vermicularis*, the pinworm
- B. *Ascaris lumbricoides*, the large intestinal roundworm
- C. *Taenia saginata*, the beef tapeworm
- D. *Schistosoma mansoni*, one of the blood flukes

Microbiology/Apply knowledge of life cycles and epidemiology/Parasitology/1

65. A helminth egg is described as having terminal polar plugs. The *most* likely helminth is:

- A. Hookworm
- B. *Trichuris trichiura*
- C. *Fasciola hepatica*
- D. *Diphyllobothrium latum*

Microbiology/Apply knowledge of organism morphology/Parasitology/1

66. Ingestion of which of the following eggs will result in infection?

- A. *Strongyloides stercoralis*
- B. *Schistosoma japonicum*
- C. *Toxocara canis*
- D. *Opisthorchis sinensis*

Microbiology/Apply knowledge of life cycles/Parasitology/2

67. *Plasmodium vivax* and *Plasmodium ovale* are similar because they:

- A. Exhibit Schüffner dots and have a true relapse in the life cycle
- B. Have no malarial pigment and multiple rings
- C. Commonly have appliqué forms in the red cells
- D. Have true stippling, do not have a relapse stage, and infect old red cells

Microbiology/Apply knowledge of life cycles and morphology/Parasitology/2

Answers to Questions 62–67

- 62. B** The coccidian *C. cayetanensis* has been linked to mini-outbreaks of diarrheal disease. Epidemiological evidence strongly implicates various berries, basil, mesclun, and snow peas as likely causes. These outbreaks are very sporadic and tend to occur primarily in March through May, primarily from imported foods.
- 63. C** The adult *O. volvulus* reside in subcutaneous nodules, and the microfilariae are found

in the fluids right under the outer layers of skin; thus the appropriate diagnostic test is the microscopic examination of skin snips for the presence of microfilariae.

64. **A** The pinworm, *E. vermicularis*, is the most common parasitic infection throughout the world, and the eggs are infective within just a few hours. It has been said, “You either had the infection as a child, have it now, or will have it again when you have children.”
65. **B** The eggs of *T. trichiura* (the whipworm) have been described as being barrel shaped, with a thick shell and two polar plugs, one at each end of the egg.
66. **C** The eggs of *T. canis* are infectious for humans and cause visceral and ocular larva migrans. These ascarid eggs of the dog can infect humans; the eggs hatch and the larvae wander through the deep tissues, occasionally the eye. In this case, the human becomes the accidental intermediate host.
67. **A** Both *P. vivax* and *P. ovale* infect young RBCs, have true stippling (Schüffner dots), contain malarial pigment, and have a true relapse stage in the life cycle.

68. The term *internal autoinfection* can be associated with the following parasites:

- A.** *Cryptosporidium* spp. and *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*)
- B.** *Cystoisospora belli* and *Strongyloides stercoralis*
- C.** *Cryptosporidium* spp. and *Strongyloides stercoralis*
- D.** *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*) and *Cystoisospora belli*

Microbiology/Apply knowledge of life cycles/Parasitology/2

69. Microsporidia have been identified as causing severe diarrhea, disseminated disease in other body sites, and ocular infections. Routes of infection have been identified as:

- A.** Ingestion
- B.** Inhalation
- C.** Direct contamination from the environment
- D.** Ingestion, inhalation, and direct contamination from the environment

Microbiology/Apply knowledge of life cycles/Parasitology/2

70. An immunocompromised patient continues to have diarrhea after repeated ova and parasites (O&P) examinations (sedimentation concentration, trichrome permanent stained smear) were reported as negative; organisms that might be responsible for the diarrhea include:

- A.** *Cryptosporidium* spp., *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*), and *Cystoisospora belli*
- B.** *Giardia lamblia* (*G. intestinalis*, *G. duodenalis*), microsporidia, and *Endolimax nana*
- C.** *Taenia solium* and *Endolimax nana*
- D.** *Cryptosporidium* spp. and microsporidia

Microbiology/Apply knowledge of life cycles and diagnostic procedures/Parasitology/3

71. Confirmation of an infection with microsporidia can be achieved by seeing:

- A.** The oocyst wall
- B.** Sporozoites within the oocyst
- C.** Evidence of the polar tubule
- D.** Organisms stained with modified acid-fast stains

Microbiology/Apply knowledge of life cycles and diagnostic procedures/Parasitology/3

72. Early ring forms of the fifth human malaria, *Plasmodium knowlesi*, resemble those of:

- A. *Plasmodium vivax*
- B. *Plasmodium ovale*
- C. *Plasmodium malariae*
- D. *Plasmodium falciparum*

Microbiology/Apply knowledge of organism morphology/Parasitology/2

73. Parasite stages that are immediately infective for humans on passage from the gastrointestinal tract include:

- A. *Schistosoma* spp. eggs
- B. *Toxoplasma gondii* bradyzoites
- C. *Ascaris lumbricoides* eggs
- D. *Cryptosporidium* spp. oocysts

Microbiology/Apply knowledge of life cycles and organism morphology/Parasitology/3

74. Older developing stages (trophozoites, schizonts) of the fifth human malaria, *Plasmodium knowlesi*, resemble those of:

- A. *Plasmodium vivax*
- B. *Plasmodium ovale*
- C. *Plasmodium malariae*
- D. *Plasmodium falciparum*

Microbiology/Apply knowledge of organism morphology/Parasitology/2

Answers to Questions 68–74

- 68. C** Both *Cryptosporidium* and *Strongyloides* have the capability for internal autoinfection in their life cycles. This means that the cycle and infection can continue even after the patient has left the endemic area. In the case of *Cryptosporidium*, the cycle continues in patients who are immunocompromised and incapable of self cure.
- 69. D** Infectious routes for microsporidial infections have been confirmed as ingestion and inhalation of the spores; direct transfer of infectious spores from environmental surfaces to the eyes has also been reported.
- 70. D** Routine O&P examinations usually do not allow the detection of *Cryptosporidium* oocysts and microsporidial spores; special stains are required. Modified acid-fast stains for *Cryptosporidium* and modified trichrome stains for the microsporidia are recommended.
- 71. C** Evidence of the cross/diagonal line polar tubules provides confirmation for microsporidial spores in the clinical specimen. Not all spores will contain visible polar tubules; the stain of choice is the modified trichrome stain (Ryan/blue or Weber/green).
- 72. D** Early ring forms of the fifth human malaria, *P. knowlesi*, resemble most closely ring forms of *P. falciparum*. However, in the early ring form stage, all human malaria species tend to look very similar.
- 73. D** On passage in stool from the gastrointestinal tract, the oocysts of *Cryptosporidium* are immediately infective; nosocomial infections have been documented with this organism. The oocysts of other coccidia have to undergo further development before they are infective.

74. C Older developing stages (developing trophozoites, schizonts) of the fifth human malaria, *P. knowlesi*, tend to closely resemble those of *P. malariae* (band forms, “daisy head” mature schizont).

75. Autofluorescence requires no stain and is recommended for the presumptive identification of:

- A. *Entamoeba histolytica* cysts
- B. *Toxoplasma gondii* tachyzoites
- C. *Dientamoeba fragilis* trophozoites
- D. *Cyclospora cayetanensis* oocysts

Microbiology/Apply knowledge of organism morphology and diagnostic procedures/Parasitology/3

76. Key characteristics of infection with *Plasmodium knowlesi* include:

- A. Rapid erythrocyte cycle (24 hours), will infect all ages of RBCs, and can cause serious disease
- B. Erythrocytic cycle limited to young RBCs and causes a relatively benign disease
- C. The possibility of a true relapse from the liver, infection in older RBCs, and causes serious disease
- D. Extended life cycle (72 hours), will infect all ages of RBCs, and disease is similar to that caused by *Plasmodium ovale*.

Microbiology/Apply knowledge of life cycles and pathogenesis/Parasitology/3

77. Microsporidial infections can be confirmed by using:

- A. Light microscopy and modified trichrome staining
- B. Phase contrast microscopy and routine trichrome staining
- C. Electron microscopy and modified acid-fast staining
- D. Fluorescence microscopy and hematoxylin staining

Microbiology/Apply knowledge of organism morphology and diagnostic procedures/Parasitology/3

78. Although the pathogenicity of *Blastocystis* spp. remains controversial, newer information suggests that:

- A. Most organisms are misdiagnosed as artifacts
- B. Approximately 10 subtypes/strains are included in the name, some of which are pathogenic and some are nonpathogenic
- C. The immune status of the host is solely responsible for symptomatic infections
- D. The number of organisms present determines pathogenicity

Microbiology/Apply knowledge of organism morphology and pathogenesis/Parasitology/3

79. Potential problems using EDTA anticoagulant and holding the blood too long prior to preparation of thick and thin blood films include:

- A. Changes in parasite morphology, loss of organisms within several hours, and poor staining
- B. Loss of Schüffner dots, poor adherence of the blood to the glass slide, and parasites beginning the vector cycle within the tube of blood
- C. None of the above
- D. All of the above

Microbiology/Apply knowledge of specimen collection and processing, organism morphology, and diagnostic procedures/Parasitology/3

Answers to Questions 75–79

75. **D** Autofluorescence is often used in preliminary testing for the presence of *C. cayetanensis* oocysts (8–10 μm). If positive, then the specimen can be stained using modified acid-fast stains for confirmation.
76. **A** Characteristics of the fifth human malaria, *P. knowlesi*, include a rapid erythrocytic cycle of 24 hours, possible infection of all ages of RBCs (as with *P. falciparum*), and the potential for serious disease (as with *P. falciparum*).
77. **A** The use of a modified trichrome stain (Ryan/blue, Weber/green; both with a very high dye content) and light microscopy (oil immersion using the 100 $\times$ objective) can be used to confirm a microsporidial infection.
78. **B** Although the pathogenicity of *Blastocystis* spp. continues to be debated (some patients are symptomatic, and some are asymptomatic), it is now known that the organisms we see microscopically comprise a number of subtypes/strains/subspecies. Approximately half these subtypes are pathogenic for humans, and half are nonpathogenic; however, morphologically they all look the same. Correct reporting of this infection should be *Blastocystis* spp.; numbers should also be reported (rare, few, moderate, many, packed).
79. **D** All of the changes mentioned are seen in a tube of EDTA blood that has been held too long prior to blood film preparation. Many morphological changes of the parasites, as well as lost organisms, occur under these conditions. These blood specimens should always be handled *STAT* (ordering, collection, processing, examination, and reporting).

80. Key characteristics of the thick film include:

- A.** The ability to see the parasite within the RBCs
 - B.** The ability to identify the parasites to the species level
 - C.** The examination of less blood than the thin blood film
 - D.** The necessity to louse the RBCs during or prior to staining
- Microbiology/Apply knowledge of specimen processing and diagnostic procedures/Parasitology/2*

Answer to Question 80

- 80. D** With thick blood films, the RBCs must be ruptured/laked either prior to or during staining. Without laking the RBCs, the stained smears would be too thick or too dark to be able to visualize any morphology. In contrast, the thin blood film is fixed prior to/during staining to preserve the RBC morphology; the parasite morphology within the infected RBC can be reviewed for possible diagnosis to the species level. The thin film has a much smaller amount of blood than the thick blood film.